

CLINICIAN UPDATE

Varicose Veins

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Case Presentation 1: A 38-year-old woman presented 6 months postpartum from the birth of her third child with pruritic bluish veins and a burning sensation behind her knees. She was embarrassed to wear skirts and dresses. On physical examination, she was obese and had prominent reticular veins behind her knees and along her lateral lower thighs (Figure 1).

Case Presentation 2: A 58-year-old woman presented with several years of painful bulging leg veins. An intensive care nurse, she noted heaviness and an aching pain that was worst at the end of her shift. On physical examination, she has large rope-like varicose veins along the right lower leg and left thigh (Figure 2).

Overview

Varicose veins are part of the spectrum of chronic venous disease and include spider telangiectasias, reticular veins, and true varicosities. Approximately 23% of US adults have varicose veins.¹ If spider telangiectasias and reticular veins are also considered, the prevalence increases to 80% of men and 85% of women.² Generally more common in women and older adults, varicose veins affect 22 million women and 11 million men between the ages of 40 to

80 years.¹ Of these, 2 million men and women will develop symptoms and signs of chronic venous insufficiency, including venous ulceration.

The sheer prevalence of varicose veins and the substantial cost of treating late complications such as chronic venous ulcers contribute to a high burden on health care resources.² Chronic venous ulcerations result in the loss of 2 million workdays and cost an estimated \$3 billion per year to treat in the United States.³ Even varicose veins alone, without more advanced signs of chronic venous insufficiency, result in important reductions in quality of life.⁴

Risk Factors, Anatomy, and Pathophysiology

Risk Factors

Risk factors for varicose veins can be categorized as hormonal, lifestyle, acquired, and inherited (Table 1). The effect of estrogen on the risk of varicose veins may explain, in part, the increased prevalence among women. Smoking is an important modifiable risk factor for varicose veins and more severe forms of chronic venous disease, including venous ulceration.⁵ Post-thrombotic syndrome after deep vein thrombosis (DVT) may result in

varicose veins in the absence of primary venous disease.

Anatomy

Venous drainage of the lower extremities is accomplished by a network of superficial veins connected to the deep veins by small perforator veins. Although disease in any of these venous systems may result in varicose veins, symptoms and their severity increase with the number of systems affected.⁶ Through a variety of pathophysiological mechanisms, weakness develops in the vein wall that results in varicosity over time. Varicosities typically form in the greater and lesser saphenous veins but also develop in branch vessels. Obstruction of the iliac veins or inferior vena cava can result in extensive varicose veins.

Pathophysiology

Venous hypertension, venous valvular incompetence, structural changes in the vein wall, inflammation, and alterations in shear stress are the major pathophysiological mechanisms resulting in varicose veins. Venous hypertension is caused by reflux attributable to venous valvular incompetence, venous outflow obstruction, or calf-muscle pump failure.² Venous reflux may occur

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Figure 1. Prominent reticular veins in the left popliteal fossa.

in either or both the superficial or deep venous system and results in venous hypertension below the area of venous valvular incompetence. In patients with perforator vein incompetence, high pressures generated in the deep veins during calf muscle contraction may be directly transmitted to the superficial

system. Valvular incompetence may result from deformation, tearing, thinning, and adhesion of the valve leaflets.

Structural changes in the vein wall contribute to pathological weakening and resultant dilation. Overproduction of collagen type I, decreased synthesis of collagen type III, and disruption of

the arrangement of smooth muscle cells and elastin fibers have been observed in histological studies of varicose venous segments.² Increased levels of tissue inhibitors of matrix metalloproteinases observed in varicose vein specimens may favor the deposition of extracellular matrix material in the vein wall. Increased levels of transforming growth factor β_1 and fibroblast growth factor β have also been observed in the walls of varicose veins and may contribute to structural degradation.

In animal models, the concentration of neutrophils, monocytes, macrophages, and lymphocytes and levels of matrix metalloproteinases increased in venous valves exposed to high pressures for prolonged periods of time.² Over time, the venous valves exposed to high pressures demonstrated adverse remodeling with decreases in leaflet length and thickness. Turbulent flow, reversal of flow, and decreases in shear stress promote inflammatory and prothrombotic changes that may further contribute to loss of structural and functional integrity of the vein wall and valve leaflets.

Clinical Presentation

Venous varicosities can be categorized according to the CEAP classification, which takes into account class (C_{1-6}), etiology (E), anatomy (A), and pathophysiology (P).⁷ Spider telangiectasias and reticular veins (C_1) describe dilated intradermal venules (<1 mm in diameter) and dilated, nonpalpable, subdermal veins (1–3 mm in diameter), respectively. True varicose veins (C_2) are “rope-like” dilated, palpable, subcutaneous veins (>3 mm in diameter). All classes of varicose veins may present substantial cosmetic concerns.

Symptoms of varicose veins vary according to their size and extent. Initial symptoms and signs localized to the areas of varicose veins include aching or throbbing discomfort, burning, pruritus, and dry irritated skin. More advanced chronic venous disease (higher CEAP class) with venous valvular incompetence manifests with symptoms and signs such as leg heaviness and fatigue, cramping, hyperpigmentation, edema,

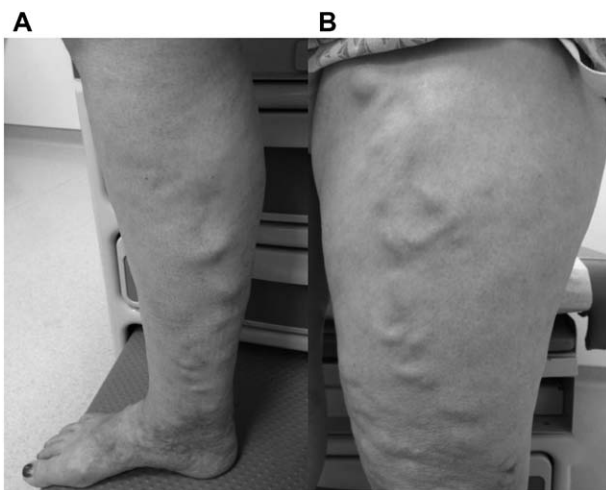


Figure 2. Large rope-like varicose veins along the right lower leg and left thigh.

Table 1. Risk Factors for Varicose Veins

Category	Risk Factor	Proposed Mechanism
Hormonal	Female gender	High estrogen state
Lifestyle	Prolonged standing and/or sitting	Venous hypertension
	Smoking	Venous endothelial injury
Acquired	Obesity	Venous hypertension
	Pregnancy	High estrogen state Venous hypertension
	Deep vein thrombosis	Deep venous obstruction Venous valvular incompetence
	Age	Venous valvular incompetence
Inherited	Family history	Venous valvular incompetence
	Tall height	Venous hypertension
	Congenital syndromes	Venous valvular incompetence Venous hypertension Deep venous obstruction

fibrotic skin changes (lipodermatosclerosis), and ulceration.

Diagnosis

Clinical Examination

The clinical evaluation of a patient with varicose veins begins with the physical examination to determine the type, location, extent, and possibly the cause of the venous disease. Varicose veins should be examined in the standing position and inspected for erythema, tenderness, or induration that may suggest superficial vein thrombosis. The clinical examination should identify any signs of more advanced chronic venous disease such as edema, hyperpigmentation, lipodermatosclerosis, atrophie blanche, or ulceration. A complete pulse examination should be performed.

The Brodie-Trendelenberg test can help distinguish between superficial venous and deep venous insufficiency and is performed with the patient recumbent, the leg elevated to 45°, and a tourniquet applied to the midthigh after the veins have completely drained. On standing, if venous refill distal to the tourniquet occurs in <30 seconds, an incompetent deep and perforator system is present. Superficial venous incompetence is present if superficial varicose veins fill rapidly on tourniquet release.

The Perthes test can distinguish between deep venous insufficiency

and obstruction and is performed in the standing position with a tourniquet applied to the midthigh. If the varicose veins collapse after a 5-minute walk, the perforator veins are competent and the deep veins are patent. If the varicose veins become more prominent and painful with walking, the deep veins are obstructed.

Role of Imaging

If the cause of varicose veins is not clear from the clinical examination or if an intervention is being considered, venous ultrasonography to evaluate for superficial and deep venous reflux should be performed. Venous reflux is defined as retrograde flow of >0.5 seconds in duration after distal manual augmentation. Venous ultrasonography can also detect deep or superficial venous thrombosis. If obstruction or extrinsic compression of iliac venous segments or the inferior vena cava is suspected, additional imaging such as computed tomographic, magnetic resonance, or invasive venography may be indicated.

Prognosis

If contributing factors are not corrected and treatment is not instituted, varicose veins may progress in severity and extent. More advanced forms of chronic venous insufficiency, including lower extremity edema and

venous ulceration, may develop and cause further decrement in quality of life and functional status. Varicose veins may thrombose or rupture and bleed, especially when large, traumatized, or located over bony prominences. In a large observational cohort study, varicose veins were associated with a 7-fold increased risk of DVT.⁸

Treatment

Selection of therapy for varicose veins should take into account symptoms, location, severity, and cause. Management options include lifestyle modifications, compression therapy, local ablative therapies, surgical interventions, and endovenous ablative therapies (Table 2). The majority of patients with varicose veins will require a multifaceted approach. In addition to symptoms refractory to noninvasive measures and cosmetic concerns, recurrent varicose vein hemorrhage and superficial thrombophlebitis are indications for invasive vein therapy. Invasive vein therapies such as sclerotherapy, surgery, or endovenous ablation are contraindicated in pregnancy, acute venous thromboembolism, and peripheral artery disease (ankle:brachial index <0.9).

Lifestyle Modification

Whether or not more advanced therapies such as ablation are considered, lifestyle modification is crucial to ensure as complete and durable a treatment response as possible. Because varicose veins are associated with obesity, weight loss is an important step in reducing progression and preventing recurrence. Regular physical activity such as walking and foot flexion exercises may improve calf muscle pump function. Elevation of the feet to at least heart level for 30 minutes at least 4 times a day and avoidance of prolonged standing and sitting decompress lower extremity veins and improve symptoms. Smoking cessation should be emphasized in patients with varicose veins.

Compression Therapy

Compression stockings are frequently prescribed as the first step in varicose

Table 2. Treatment Options for Varicose Veins

Intervention	Type of Varicose Vein
Lifestyle modification Weight loss Exercise Elevation Avoid prolonged standing/sitting Smoking cessation	All types
Compression therapy 15–20 mm Hg 20–30 mm Hg 30–40 mm Hg	Minor varicosities and symptoms Moderate-to-severe varicosities and symptoms Severe varicosities with chronic venous insufficiency
Local ablative therapy Sclerotherapy, thermocoagulation, or cutaneous laser	Spider telangiectasias, reticular veins
Surgery Stab/microincision phlebectomy Surgical stripping	Large branch or residual postablation varicosities Saphenous varicosities
Endovenous ablation Radiofrequency or laser	Saphenous varicosities

vein management and are effective for treatment of discomfort and edema.¹ Compression stockings improve venous hemodynamics by decreasing venous reflux and reducing ambulatory venous hypertension. Because of limited randomized, controlled trial evidence, the impact of compression stockings on progression or recurrence of varicose veins remains unclear.⁹ A multicenter randomized, controlled trial of active versus placebo compression stockings suggested that routine use of compression stockings may not prevent post-thrombotic syndrome in patients with first proximal DVT.¹⁰

Patients should be instructed to don compression stockings in the morning while the legs are in a nondependent position and to remove them at night before going to bed. Increasing compression strength is prescribed to treat larger varicosities and greater severity of symptoms and chronic venous insufficiency. Compression stocking length should cover all areas affected by varicose veins. Unfortunately, the rate of nonadherence to compression stocking regimens approaches 60% in patients with chronic venous disease, including varicose veins.¹¹ Compression stockings may not be practical for elderly patients, morbidly obese patients, those

with cellulitis or active ulceration, and those with peripheral artery disease. Devices to assist putting on compression stockings are available and may improve adherence in patients with difficulty applying their stockings.

Although most payers require a trial of compression stockings (often 3 months) before providing coverage for more invasive therapies, clinical practice guidelines from the Society for Vascular Surgery and the American Venous Forum recommend against compression therapy as the primary treatment of symptomatic varicose veins in patients who are candidates for endovenous ablation.¹²

Therapies for Telangiectasias and Reticular Veins

Patients with symptomatic or cosmetically bothersome spider telangiectasias, reticular veins, and some small varicose veins may be treated effectively with a number of local ablative therapies, including sclerotherapy, thermocoagulation, and cutaneous laser. Each technique relies on endothelial injury, either chemical- or heat-based, that results in thrombosis and eventual fibrosis of the veins. Providers should emphasize to patients that most will require multiple treatments and

that most payers will not cover vein therapies for cosmetic indications.

Complications of sclerotherapy include allergic reactions to sclerosants, hyperpigmentation (especially if performed during months of high sun exposure), superficial thread-like capillaries causing a blush discoloration (called matting), cellulitis, and rarely ulceration or thromboembolism. Patients may require additional office visits to treat areas of trapped coagulum. Thermocoagulation and cutaneous laser therapy may result in skin damage as well as hypo- or hyperpigmentation.

Sclerotherapy results in cosmetic improvement in 70% of patients and patient satisfaction in excess of 70%.¹³ Best results are achieved when compression stockings are worn over treated areas for 7 to 10 days after sclerotherapy.¹⁴ Randomized, controlled data evaluating thermocoagulation and cutaneous laser therapy are limited. Cutaneous laser therapy is useful in patients who wish to avoid needles.

Clusters of varicose veins in absence of venous reflux suggest perforator insufficiency. Ultrasound-guided sclerotherapy and ablation using special endovenous probes have been used to treat perforator vein reflux.

Therapies for Varicose Veins

Varicosities in branches of the major superficial veins can be treated using stab or microincision phlebectomy, which requires only local tumescent anesthesia and leaves minimal scars. Tumescent anesthesia involves injection of large volumes of a local anesthetic solution. These microsurgical techniques have replaced more traditional large-incision phlebectomy.

For varicosities in the greater and lesser saphenous veins, endovenous techniques have largely replaced traditional large-incision surgical stripping and vein ligation. Surgical stripping is associated with varicose vein recurrence in up to 50% of patients by 5 years, most often as a result of incomplete phlebectomy, persistent venous reflux, or neovascularization.¹ Complications of surgical stripping include extensive

ecchymosis and scarring, hematoma, lymphocele, infection, nerve injury, and DVT. Surgical stripping and endovenous ablation for saphenous vein varicosities should be avoided in patients with deep venous obstruction or deep venous reflux because of an increased risk of venous ulceration.

Endovenous therapy with either radiofrequency or laser ablation relies on thermal injury to cause thrombotic and fibrotic closure of the saphenous veins in patients with documented superficial venous reflux. Local tumescent anesthesia is required for patient comfort, to separate the vein from the skin surface, and to protect surrounding tissue from heat injury. A systematic review demonstrated that 3-year estimated pooled success rates of endovenous therapies for saphenous varicosities were 84% for radiofrequency ablation and 94% for endovenous laser ablation compared with 78% for surgical stripping.¹⁵ Long-term randomized, controlled trial data have established endovenous therapies as effective and durable alternatives to surgical stripping.^{16,17} Utilization of endovenous therapies for varicose veins has grown as a result of patient preference for less invasive outpatient procedures, favorable reimbursement, and the wider range of clinicians able to offer these techniques.

In a randomized, controlled trial comparing endovenous laser ablation and surgical stripping for greater saphenous varicose veins, 98% of patients were symptom free at 1 year in both groups.¹⁸ Compared with surgical stripping, endovenous ablation offers advantages of faster return to physical activity and work and shorter length of disability.¹⁹ In an economic analysis of varicose vein therapies, the estimated direct cost of performing endovenous radiofrequency ablation in a treatment room was \$906 compared with an estimated \$4241 for surgical stripping in an operating room followed by in-hospital observation.²⁰

Recurrence of varicose veins may follow endovenous ablation as a result of recanalization of the saphenous veins. Return visits for microincision

phlebectomy may be required to treat residual or recurrent varicose veins after ablation. Surgical stripping is preferred over endovenous ablation after a failed attempt at endovenous therapy and when the greater saphenous vein is too tortuous, aneurysmal, or close to the skin surface. Complications of endovenous ablation include ecchymosis, hematoma, skin burns, DVT (<3%), and nerve injury.⁶

Typically, endovenous therapy is performed in 1 leg at a time. In patients with spider telangiectasias or reticular veins in addition to larger varicose veins, treatment of saphenous venous varicosities should be performed first because relieving venous reflux may resolve or improve these superficial varicosities, thereby reducing the need for subsequent local therapies.

Foam sclerotherapy is an emerging technique that may be useful for treating varicose veins ranging from spider telangiectasias to saphenous varicosities.²¹ Mixing air with conventional sclerosants allows for great contact area with vein walls, even in larger varicose veins. Concerns regarding embolization of foam sclerosant during saphenous vein ablation in earlier reports have limited enthusiasm for this technique.

Case Disposition 1: Because of her symptoms and the cosmetic impact of her reticular veins, she was offered sclerotherapy. After three serial sessions of sclerotherapy, she had near-complete resolution of her reticular veins and the associated burning sensation.

Case Disposition 2: Venous ultrasonography was remarkable for superficial venous reflux in the greater saphenous veins bilaterally. The patient was prescribed a 3-month trial of 30- to 40-mmHg grade graduated compression stockings with minimal relief of symptoms. She subsequently underwent staged endovenous laser ablation of the left leg first and then the right leg. She had an excellent cosmetic result with complete resolution of her symptoms. She was advised to wear thigh-high 20- to 30-mmHg graduated compression stockings to prevent recurrence.

Disclosures

None.

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